Arm Valuation

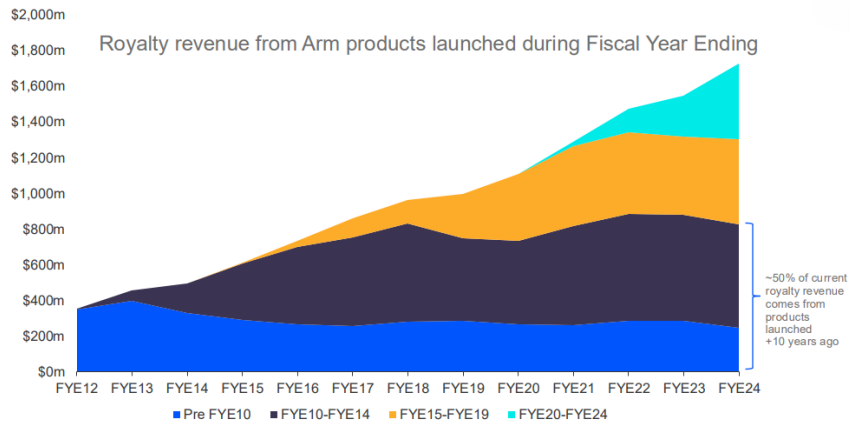
Salvatore Gabriele Messina

1 The Story

Arm Holdings plc, based in Cambridge, UK, is an engineer-led company specializing in semiconductor intellectual property (IP) with a focus on CPU, GPU, and AI architectures. Its R&D-driven culture enables continuous innovation, allowing Arm to license IP to mobile OEMs, embedded systems and the growing data-center and AI-server markets. The company went public in September 2023 but has been profitable for many years.

Arm operates an asset-light model, generating high-margin cash flows through license fees, royalties, and custom agreements. With minimal fixed capital requirements, low effective tax rates, and targeted government incentives, the company maintains efficient operations.

Growth is fueled not only by rising partner volumes, new market segments and IP advancements but also by the reuse of existing chip designs. Architectures licensed today are often repurposed in consumer electronics or industrial applications, where designing new chips is not cost-effective. This reuse extends the royalty stream well beyond 20 years, while total processor volume growth offsets natural revenue decay.



Arm's Neoverse platform demonstrates its move into AI inferencing, diversifying revenue into high-value workloads. Because building custom AI server chips is expensive, demand for Arm's licensed IP is likely to persist. As models become more efficient, the reuse dynamics observed in consumer markets may also apply to AI.

A notable factor is Arm's relationship with Arm China, a separate entity that contributes roughly 20% of total revenues. Significant revenue also comes from Taiwan and Korea. Less than half of total revenues originate in the Americas, increasing geopolitical exposure. This is reflected in our 2025 equity-risk-premium (ERP) assumptions, which remain consistent with those for 2024. Additionally, over 50% of revenue comes from just five customers, including Arm China [2].

Arm is not just a patent repository; it is built around talent. Its success depends heavily on hiring top-tier engineers to develop cutting-edge IP. The competitive landscape is intense, and innovation is increasingly difficult. There is a material risk that Arm could lose its lead in high-value designs, whose long-tail royalties are central to its business model. Market dominance is not guaranteed, as past industry shifts (e.g., Nokia, Olivetti) have shown.

To attract top engineering talent, Arm relies in part on stock-based compensation. This creates dilution, which is expected to continue, as competitive bonuses remain necessary.

Capitalizing R&D increases reported invested capital and reduces adjusted ROIC in the short term. However, these investments are crucial to sustaining Arm's growth and innovation pipeline.

Arm's royalty-based structure makes it well suited for a DCF approach. Relative valuation is less reliable due to the company's unique model and the absence of close comparables. However, given Arm's cyclicality and exposure to a volatile sector, an option-like valuation approach is warranted. A Monte Carlo simulation has therefore been implemented, modeling cash flows as a geometric Brownian motion with drift.

Finally, while Arm does not currently manufacture chips, it retains the strategic option to do so. This latent capability introduces optionality that could be valuable in an M&A context—though no such deal is currently confirmed, and any potential synergy remains speculative at this stage.

2 Intrinsic valuation

An analysis was conducted across three periods: a growth period, a convergence period, and a perpetuity period for the terminal value. The study explored how varying parameters—such as the duration of growth and the operating margin (OM)—affect valuation. Additionally, a Monte Carlo analysis was performed to simulate the variability of revenues.

Based on the intrinsic valuation, even under broadly optimistic assumptions, the company appears **overvalued** in all cases. I even ran a Monte Carlo simulation with OM fixed at 0.7 (7-year), and even in that case, the 99th percentile value—111.81—remains below the current market price.

If I had to select the assumption I consider most reasonable, I would choose 45.43, the 50th percentile of the 7-year Monte Carlo distribution (the median). This reflects a reasonable OM as well as an adjusted OMAT declining toward 35%, with an expected ROIC stabilizing around 33%. Under this assumption, the company appears overvalued by $\frac{133.07}{45.43} - 1 \approx 193\%$. (ARM's price is as of market close on 15 May.)

It is worth noting that if I had assumed a higher WACC—while maintaining a 2% spread in the terminal ROIC—I would have obtained a higher terminal value, whose present value would weigh as much as the entire sum of the 10-year FCFF.

For more details on the exponential convergence formula and the calculation of R&D see Appendix. The Python code used for the analysis is referenced in the bibliography [4].

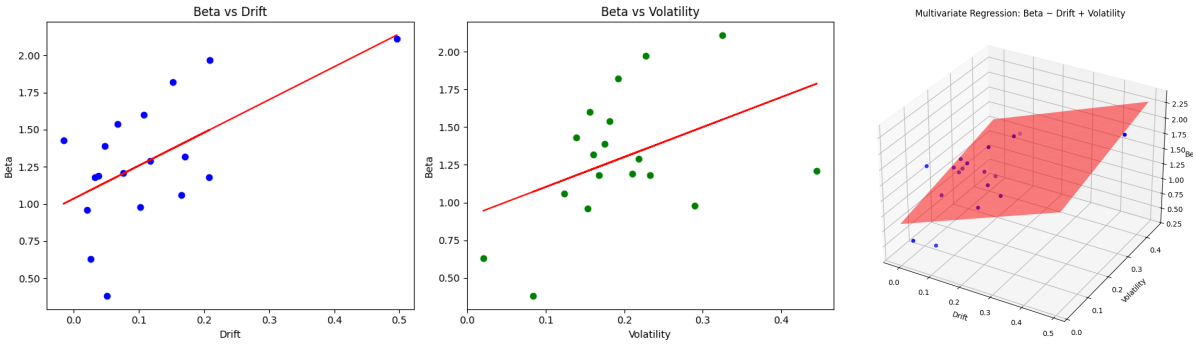
2.1 Calculation of WACC

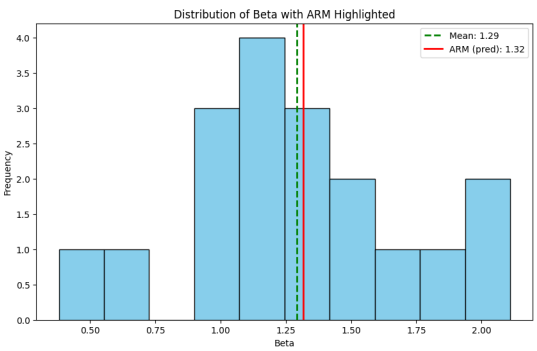
ERP. To calculate the equity risk premium, we took a revenue-weighted average of regional ERPs based on Arm's revenue mix. The weighted average ERP is:

$$\text{ERP}_{\text{WA}} = 6.31\%.$$

Country/Region	% Revenue	ERP (%)
USA	43.7%	4.33
China	21.6%	5.27
Taiwan	16.1%	5.13
South Korea	9.5%	20.35
Others	9.1%	5.76

Beta Estimation. Being a cyclical company, we wanted ARM's beta to reflect the variance of its cash flows. By selecting the top 20 U.S.-listed semiconductor peers and computing log-returns on their last five annual revenues, we estimated drift μ_i and volatility σ_i; applying simple OLS regressions yielded $\beta_{\text{drift}} \approx 1.41$ and $\beta_{\text{volatility}} \approx 1.15$, while the multivariate model produced $\beta_{\text{multi}} \approx 1.3175$. On balance, we adopt a final $\beta = 1.3175$ for ARM's CAPM-based cost of equity.





ARM's chosen $\beta = 1.3175$ vs sample distribution

	Value	p-val	R^2	Adj. R^2
Drift model				
Intercept	1.0361	0.0000	0.3560	0.3150
Drift	2.2178	0.0090	0.3560	0.3150
Volatility model				
Intercept	0.9052	0.0010	0.1840	0.1330
Volatility	1.9832	0.0753	0.1840	0.1330
Multivariate model				
Intercept	0.8494	0.0010	0.4100	0.3320
Drift	1.8893	0.0300	0.4100	0.3320
Volatility	1.1534	0.2570	0.4100	0.3320
ARM parameter estimates				
Drift	0.1704			
Volatility	0.1268			

Regression results and ARM's estimated parameters

Arm's total financial debt consists entirely of capitalized lease liabilities of approximately $346 million, for which the implicit coupon rate is 2.85%.

We estimate the cost of equity as $r_e = 4.43\% + 1.3175 \cdot 6.31\% = 12.69\%$, and the after-tax cost of debt as $r_d(1-t) = 2.85\% \cdot (1-0.10) = 2.565\%$. Using book values of equity at $6,839m and debt at $346m, leading to a WACC of approximately $0.9517 \cdot 12.69\% + 0.0483 \cdot 2.565\% = 12.23\%$.

In the converging state, we assume a WACC of 10%, reflecting a lower beta and a reduced equity risk premium. The risk-free rate $r_f = 4,43\%$ is as of 15 May close.

2.2 Revenue Growth & Volatility

Forecasting Arm's revenue is challenging due to industry cyclicality and geopolitical risks. We model two high-growth phases (5- and 7-year) before settling into a steady state at the risk-free rate. Historical regression over the past five years shows revenue volatility of 12.67%, so we layer a Monte Carlo: volatility remains at 12.67% during high growth and converges to 1.77% in steady state (IBM proxy).

2.3 Reinvestment, Margins & Dilution

Arm settles much of its operating expenses via stock-based compensation: in 2024 its share count rose from 1.044 bn to 1.066 bn. With €4 bn revenue and a €122.4 bn market cap ($P/S \approx 30.6\times$), the implied share-paid revenue is $(1.066/1.044 - 1) \times 30.6 \approx 64.6\%$, yielding $g_{\text{share}} = 64.6\%/30.6 \approx 2.11\%$. As the company matures, both %rev both P/S and are expected to decline, but their relative speed is uncertain—making the overall effect on dilution difficult to predict. For simplicity we model dilution at 3% of revenue converging to 1%. $E_t = E_{t-1} + dil_t \cdot Rev_t$.

CapEx, depreciation and working-capital changes are negligible, making EBIT and OM (50%) central. We consider three margin scenarios: (1) maintain 50 % during the growth phase then converge to 30 %, (2) hold 50 % throughout both growth and steady state, and (3) begin below 50 % and rise to a higher long-run margin. All three are plausible: (1) captures scale-driven leverage, (3) models margin tailwinds from reduced R & D spend, and (2) blends both effects. We report results for all scenarios and invite the reader to choose according to his preference on expected ROIC and it's evolution over time; our Monte Carlo uses scenario (2).

2.4 FCFF & R&D Capitalization

Under , $\text{CapEx} - \text{Depreciation} + \Delta\text{NCWC} = 0$, free cash flow is $FCFF = EBIT(1-t)$, since R&D capitalization does not affect FCFF. We assume a ten-year R&D life, amortizing as $AmortRD_t = \frac{1}{10}\sum_{i=1}^{10} RD_{t-i}$ and capitalizing as $CapRD_t = \frac{1}{10}\sum_{i=0}^{10}(10-i)\,RD_{t-i}$. The R&D/revenue ratio starts at $p_0 = 0.5(1-OM)$ and converges to $p^* = \min\{5\%, 0.1(1-OM)\}$ via the converging formula, so $RD_t = p_t\,Rev_t$. Invested capital at the end of the year is $IC_t = E_t + CapRD_t + Debt - Cash$.

2.5 Adjusted NOPAT & ROIC

To capture R&D's full economic cost we define $AdjNOPAT_t = Rev_t \cdot OM\,(1-tax_t) + (RD_t - AmortRD_t)$ and $AdjROIC_t = \frac{AdjNOPAT_t}{IC_{t-1}} \times 100\%$, benchmarking against market norms and using the terminal rule $FCFF = NOPAT(1 - g/ROIC_{term})$ with $ROIC_{term}$= 0.12, 2% more of terminal wacc = 0.1.

2.6 Projection Outputs and Valuation Summary

The following tables report the full FCFF projections under two scenarios: a 7-year growth period and a shorter 5-year variant, both followed by a convergence phase and terminal value estimation. Key drivers such as revenue growth, volatility, operating margins, and discount factors are explicitly shown. Table 3 compares the resulting equity values and price per share across the two models.

Table 1: FCFF Projection Transposed 7-Year Growth – fixed OM = 0.5

	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
Revenue	4.6882	5.4852	6.4177	7.5087	8.7851	10.2786	12.0260	13.3146	14.3228	15.1824	15.8549
Growth	0.1700	0.1700	0.1700	0.1700	0.1700	0.1700	0.1700	0.1072	0.0757	0.0600	0.0443
Volatility	0.1270	0.1270	0.1270	0.1270	0.1270	0.1270	0.1270	0.0724	0.0450	0.0314	0.0177
OM	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
Tax Rate	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000	0.1000	0.1500	0.1500	0.1500	0.1500
NOPAT	2.1097	2.4683	2.8879	3.3789	3.9533	4.6254	5.4117	5.6587	6.0872	6.4525	6.7383
WACC	0.1225	0.1225	0.1225	0.1225	0.1225	0.1225	0.1225	0.1169	0.1127	0.1095	0.1000
Discount F.	0.8908	0.7936	0.7070	0.6298	0.5611	0.4998	0.4453	0.3987	0.3583	0.3229	0.2936
PV FCFF	1.8794	1.9589	2.0417	2.1280	2.2180	2.3118	2.4096	2.2558	2.1809	2.0836	1.9781

Table 2: FCFF Projection Transposed 5-Year Growth – fixed OM = 0.5

	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
Revenue	4.6882	5.4852	6.4177	7.5087	8.7851	9.7265	10.4630	11.0909	11.6694	12.2322	12.7741
Growth	0.1700	0.1700	0.1700	0.1700	0.1700	0.1072	0.0757	0.0600	0.0522	0.0482	0.0443
Volatility	0.1270	0.1270	0.1270	0.1270	0.1270	0.0724	0.0450	0.0314	0.0245	0.0211	0.0177
OM	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
Tax Rate	0.1000	0.1000	0.1000	0.1000	0.1000	0.1500	0.1500	0.1500	0.1500	0.1500	0.1500
NOPAT	2.1097	2.4683	2.8879	3.3789	3.9533	4.1337	4.4468	4.7136	4.9595	5.1987	5.4290
WACC	0.1225	0.1225	0.1225	0.1225	0.1225	0.1169	0.1127	0.1095	0.1071	0.1053	0.1000
Discount F.	0.8908	0.7936	0.7070	0.6298	0.5611	0.5023	0.4515	0.4069	0.3675	0.3325	0.3023
PV FCFF	1.8794	1.9589	2.0417	2.1280	2.2180	2.0765	2.0076	1.9180	1.8228	1.7286	1.6410

Tables 4 and 5 reflect the R&D capitalization and amortization adjustments for the scenarios shown in Tables 1 and 2; immediately afterward, we present the Monte Carlo analysis for those same cases—in this context, revenue volatility, typically seen as a risk, may instead be a feature rather than a bug, potentially adding value in a high-growth industry. Finally, Table 6 shows the implied prices and ROICs under the various OM assumptions. A few duplicate cases were included to verify the code's functionality (e.g. OM 0.5 converging to 0.5 and OM 0.3 converging to 0.3).

Table 3: Valuation Summary Comparison

Metric	**7-Year**	**5-Year**
PV(FCFF 2026-2036)	23.4460	21.4205
PV(Terminal 2037)	25.0224	20.7584
Cash	2.8250	2.8250
Debt	0.3460	0.3460
Market Value of Equity	50.9473	44.6579
Total Shares (B)	1.0660	1.0660
Equity Value per Share	47.79	41.89

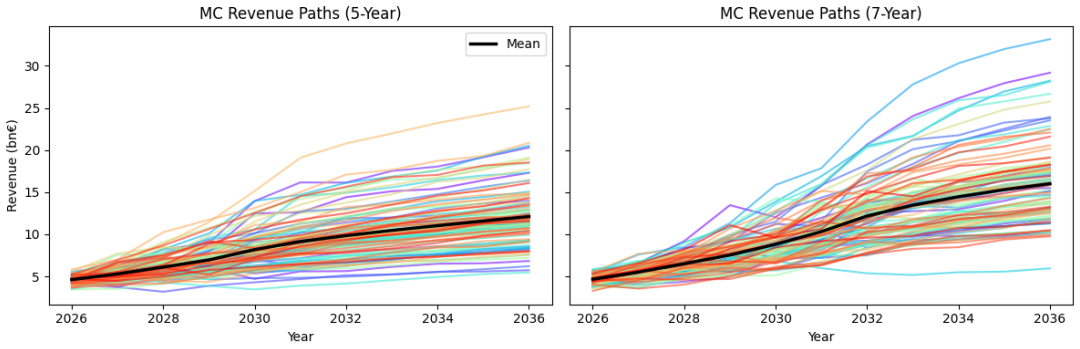
Table 4: R&D Adjustments, fixed OM = 0.5, 7-Year Growth

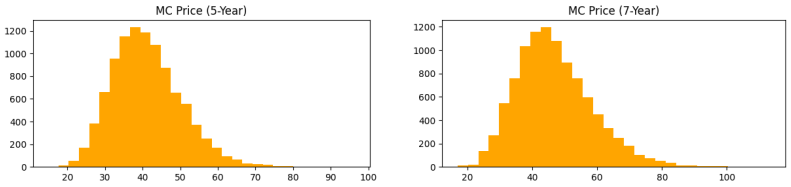
	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
NOPAT base	0.9030	2.1097	2.4683	2.8879	3.3789	3.9533	4.6254	5.4117	5.6587	6.0872	6.4525	6.7383
R&D Outlay	2.0710	1.1720	1.3713	1.6044	1.8772	2.1963	2.5697	3.0065	1.9972	1.4323	1.1387	0.7927
Adj NOPAT	2.4819	2.5825	3.0232	3.5388	4.1421	4.8479	5.6737	6.7213	5.7578	5.5350	5.6614	5.6945
Adj ROIC (%)	0.3541	25.0649	27.6931	30.4745	33.3291	36.2228	39.1197	42.4986	33.0278	30.9336	32.1536	33.4362
adj OMAT (%)	0.6194	55.0859	55.1162	55.1420	55.1641	55.1830	55.1992	55.8898	43.2447	38.6448	37.2893	35.9165
Dilution (%)	2.0700	3.0000	3.0000	3.0000	3.0000	3.0000	3.0000	3.0000	2.0000	1.5000	1.2500	1.0000
CapRD	5.9434	6.4162	6.9711	7.6220	8.3852	9.2798	10.3281	11.6377	11.7369	11.1847	10.3936	9.3498
Amortized	0.4921	0.6992	0.8164	0.9535	1.1140	1.3017	1.5213	1.6969	1.8980	1.9845	1.9298	1.8366
Book Equity	6.8390	6.9796	7.1203	7.2848	7.4774	7.7026	7.9662	8.2745	8.6353	8.9016	9.1165	9.3062
IC	10.3034	10.9169	11.6124	12.4279	13.3836	14.5034	15.8153	17.4333	17.8932	17.6073	17.0311	16.1771

Table 5: R&D Adjustments, fixed OM = 0.5, 5-Year Growth

	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
NOPAT base	0.9030	2.1097	2.4683	2.8879	3.3789	3.9533	4.1337	4.4468	4.7136	4.9595	5.1987	5.4290
R&D Outlay	2.0710	1.1720	1.3713	1.6044	1.8772	2.1963	1.4590	1.0463	0.8318	0.7293	0.6881	0.6387
Adj NOPAT	2.4819	2.5825	3.0232	3.5388	4.1421	4.8479	4.0714	3.9073	3.9545	4.1280	4.4509	4.7701
Adj ROIC (%)	0.3541	25.0649	27.6931	30.4745	33.3291	36.2228	28.0719	26.5716	27.5390	30.0054	34.0681	38.3200
adj OMAT (%)	0.6194	55.0859	55.1162	55.1420	55.1641	55.1830	41.8590	37.3436	35.6554	35.3746	36.3866	37.3421
Dilution (%)	2.0700	3.0000	3.0000	3.0000	3.0000	3.0000	2.0000	1.5000	1.2500	1.1250	1.0625	1.0000
CapRD	5.9434	6.4162	6.9711	7.6220	8.3852	9.2798	9.2175	8.6779	7.9188	7.0873	6.3395	5.6806
Amortized	0.4921	0.6992	0.8164	0.9535	1.1140	1.3017	1.5213	1.5858	1.5909	1.5608	1.4359	1.2976
Book Equity	6.8390	6.9796	7.1203	7.2848	7.4774	7.7026	7.9662	8.1607	8.3177	8.4563	8.5876	8.7175
IC	10.3034	10.9169	11.6124	12.4279	13.3836	14.5034	14.7046	14.3597	13.7575	13.0646	12.4481	11.9192

In 2025, the company reported an EBIT of 831 million, which generated a tax benefit of 72 million. As of year-end 2024, the book value of equity stood at 5.295 billion, cash at 2.923 billion, debt at 0.271 billion, and capitalized R&D at 4.3645 billion (computed as $1.9790+0.9\cdot1.1330+0.8\cdot0.9950+0.7\cdot0.8140$), resulting in invested capital of 7.0075 billion.





Quantile	5-Year	7-Year
1%	23.61	25.46
5%	27.68	30.27
10%	30.26	33.00
25%	34.57	38.57
50%	40.17	45.43
75%	46.65	53.81
90%	53.16	62.58
95%	57.51	68.32
99%	65.90	80.07

Table 6: Price/Share and ROICs Scenarios under Different OM Paths

Fixed OM

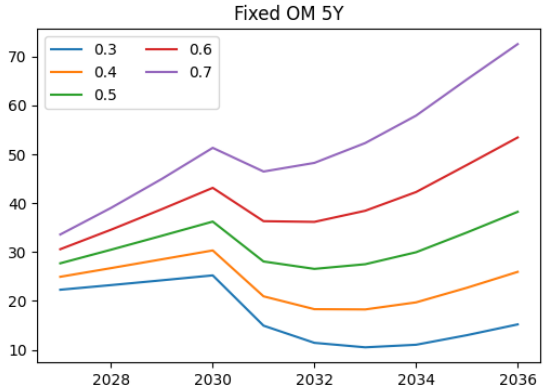
OM	5-Year	7-Year
0.30	26.07	29.61
0.40	33.98	38.70
0.50	41.89	47.79
0.60	49.81	56.89
0.70	57.72	65.98

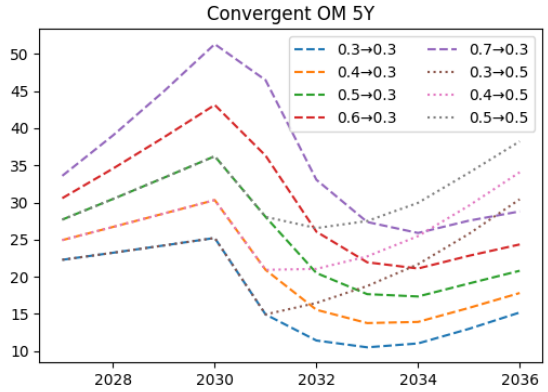
OM → 0.3

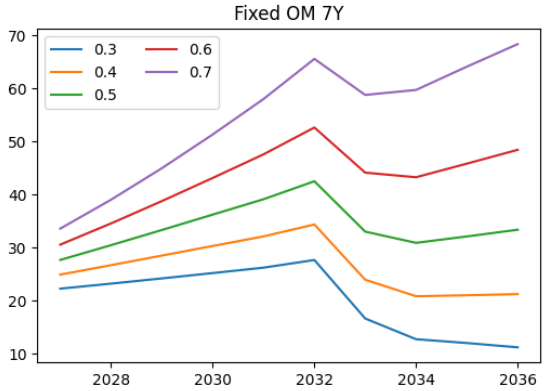
OM	5-Year	7-Year
0.30	26.07	29.61
0.40	28.72	33.14
0.50	31.36	36.67
0.60	34.01	40.20
0.70	36.66	43.73

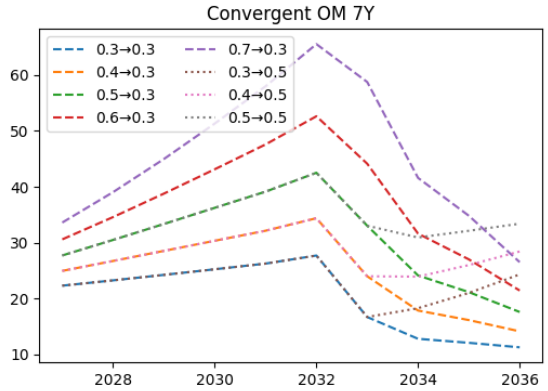
OM → 0.5

OM	5-Year	7-Year
0.30	36.59	40.73
0.40	39.24	44.26
0.50	41.89	47.79









3 Relative valuation

For relative valuation, we use *Price-to-Sales* (P/S) and *EV/Sales*. P/S divides equity value by total revenues—mismatching claims on the firm—whereas EV/Sales aligns enterprise value with the full revenue stream. Despite its *inconsistency*, P/S is widespread in technology and retail because it offers broader sample coverage (few firms report negative revenues) and revenue figures are perceived as more reliable than earnings or book value.

Intrinsic Drivers of Revenue Multiples From valuation theory, an EV/Sales multiple is driven by four levers: (1) cost of capital (↑ → lower multiple), (2) growth rate (↑ → higher multiple), (3) reinvestment rate (↑ → lower multiple), (4) operating margin (↑ → higher multiple). In practice, *Adjusted OMAT* and *Adjusted ROIC* summarize reinvestment efficiency and profitability. We capitalized R&D over five years (ten years for ARM).

Ticker	Adj. ROIC	Adj. OMAT	Drift	Volatility	P/S	EV/S	Beta
NVDA	1.6271	0.5972	0.4962	0.3246	25.2926	25.0402	2.1100
AVGO	0.1197	0.3218	0.1651	0.1235	20.0153	21.0656	1.0600
TSM	0.3691	0.4079	0.1705	0.1609	8.2482	7.7018	1.3200
QCOM	0.2296	0.2897	0.1172	0.2186	3.9694	3.9877	1.2900
AMD	0.0640	0.1626	0.2089	0.2277	6.8783	6.7853	1.9700
TXN	0.1796	0.3254	0.0208	0.1529	10.6046	11.0933	0.9600
ARM	0.3541	0.6194	0.1704	0.1268	35.2026	34.5839	1.3175
ADI	0.0422	0.2051	0.1021	0.2901	12.0478	12.5723	0.9800
MU	0.0864	0.1668	0.0759	0.4453	3.4009	3.6183	1.2100
MRVL	0.0212	0.0850	0.1518	0.1917	9.8705	10.4453	1.8200
IFX	0.0902	0.1578	0.1071	0.1563	2.7928	3.0800	1.6000
MPWR	0.2268	0.3012	0.2078	0.1674	14.7299	14.3072	1.1800
STMPA	0.0527	0.0879	0.0374	0.2106	1.6874	1.4279	1.1900
GFS	0.0694	0.1142	0.0671	0.1810	3.2616	3.0666	1.5400
ASEKY (7259)	-0.0186	-0.0126	0.0515	0.0842	0.2847	0.3349	0.3800
UMC (6615)	0.0467	0.0162	-0.0159	0.1391	0.0670	0.2156	1.4300
ON	0.1036	0.1755	0.0475	0.1750	2.8338	2.9299	1.3900
SWKS	0.0832	0.1636	0.0327	0.2327	2.7441	2.6690	1.1800
IBM	0.1128	0.1649	0.0260	0.0177	3.7315	4.5173	0.6300

In OLS regression, the Gauss–Markov theorem requires that the residuals be uncorrelated with the predictors; adding variables helps satisfy this orthogonality condition but reduces the adjusted R^2, which penalizes the inclusion of regressors. By default, OLS also assumes homoscedastic residuals—hence our heteroscedasticity tests and the switch to HC1 robust standard errors when indicated. With only $N = 18$ observations, using normal quantiles may misstate coverage; however, in the presence of heteroscedasticity one cannot validly use Student's t quantiles with degrees of freedom adjusted for regressors.

Accordingly, I used the most appropriate type of regression and quantiles based on the Breusch–Pagan test, using a p-value threshold of 0.05 to assess the behavior of the residuals.

Model	R^2	Adj. R^2
P/S ∼ ROIC + OMAT + Beta	0.647	0.572
P/S ∼ ROIC + OMAT + Drift + Volatility	0.767	0.695
EV/S ∼ ROIC + OMAT + Beta	0.616	0.534
EV/S ∼ ROIC + OMAT + Drift + Volatility	0.734	0.652

Implied P/S Both implied multiples lie well below ARM's market P/S of 35.20x, suggesting the stock is priced at a premium of roughly 75%.

$$\widehat{\mathrm{P/S}} = -0.9996 + 1.5741\,\mathrm{ROIC} + 34.1872\,\mathrm{OMAT} + 0.7489\,\beta = 21.72.$$

$$\widehat{\mathrm{P/S}} = -0.6338 - 6.2993\,\mathrm{ROIC} + 27.2280\,\mathrm{OMAT} + 41.2045\,\mathrm{Drift} - 5.9694\,\mathrm{Volatility} = 20.27.$$

Implied EV/Sales Both implied EV/Sales multiples also lie well below ARM's market EV/Sales of 34.58, suggesting the stock is priced at a premium of roughly 75%.

$$\widehat{\mathrm{EV/S}} = -0.6285 + 0.5453\,\beta + 34.6832\,\mathrm{OMAT} + 1.1742\,\mathrm{ROIC} = 21.99.$$

$$\widehat{\mathrm{EV/S}} = -0.3818 + 27.8668\,\mathrm{OMAT} - 6.7418\,\mathrm{ROIC} + 41.0006\,\mathrm{Drift} - 6.6968\,\mathrm{Volatility} = 20.63.$$

P/S	coef	std err	t	$P > \|t\|$	[0.025	0.975]	Stat
const	-0.9996	4.2890	-0.233	0.819	-10.1980	8.1980	R^2
Adjusted_ROIC	1.5741	5.4450	0.289	0.777	-10.1042	13.2525	0.647
Adjusted_OMAT	34.1872	12.7193	2.688	0.018	6.9071	61.4673	Adj. R^2
Beta	0.7489	2.9217	0.256	0.801	-5.5176	7.0154	0.572

Table 7: P/S Breusch–Pagan $p = 0.1116$ (homoscedasticity; OLS + t-CI).

P/S	coef	std err	t	$P > \|t\|$	[0.025	0.975]	Stat
const	-0.6338	1.3580	-0.467	0.648	-3.5680	2.3000	R^2
Adjusted_ROIC	-6.2993	5.5952	-1.126	0.281	-18.3870	5.7880	0.767
Adjusted_OMAT	27.2280	13.2269	2.059	0.060	-1.3470	55.8030	Adj. R^2
Drift	41.2045	15.8118	2.606	0.022	7.0450	75.3640	0.695
Volatility	-5.9694	7.3362	-0.814	0.430	-21.8180	9.8800	

Table 8: P/S Breusch–Pagan $p = 0.0419$ (heteroscedasticity; robust HC1 + normal-CI).

EV/S	coef	std err	t	$P > \|t\|$	[0.025	0.975]	Stat
const	-0.6285	4.5150	-0.139	0.891	-10.3130	9.0560	R^2
Beta	0.5453	3.0760	0.177	0.862	-6.0520	7.1430	0.616
Adjusted_OMAT	34.6832	13.3920	2.590	0.021	5.9610	63.4050	Adj. R^2
Adjusted_ROIC	1.1742	5.7330	0.205	0.841	-11.1210	13.4700	0.534

Table 9: EV/S Breusch–Pagan $p = 0.1284$ (homoscedasticity; OLS + t-CI).

EV/S	coef	std err	t	$P > \|t\|$	[0.025	0.975]	Stat
const	-0.3818	1.4220	-0.268	0.793	-3.4550	2.6910	R^2
Adjusted_OMAT	27.8668	14.5550	1.915	0.078	-3.5780	59.3120	0.734
Adjusted_ROIC	-6.7418	6.1220	-1.101	0.291	-19.9680	6.4850	Adj. R^2
Drift	41.0006	17.0140	2.410	0.032	4.2440	77.7570	0.652
Volatility	-6.6968	7.9570	-0.842	0.415	-23.8860	10.4930	

Table 10: EV/S Breusch–Pagan $p = 0.0424$ (heteroscedasticity; HC1 + normal-CI).

The correlation matrix reveals a high degree of multicollinearity between some predictors, particularly between $AdjROIC$ and $AdjOMAT$ (0.80), as well as between $AdjROIC$ and Drift (0.85). These strong correlations suggest potential redundancy in explanatory variables, which could inflate standard errors in regression models. Volatility appears to be less correlated with the other predictors, while Beta shows only moderate correlation with Drift (0.60) and $AdjROIC$ (0.46). The near-perfect correlation between P/S and EV/S (0.9989) indicates that these valuation metrics are essentially interchangeable in this dataset.

	Adj_ROIC	Adj_OMAT	Drift	Volatility	P/S	EV/S	Beta
Adjusted_ROIC	1.0000	–	–	–	–	–	–
Adjusted_OMAT	0.8024	1.0000	–	–	–	–	–
Drift	0.8479	0.7531	1.0000	–	–	–	–
Volatility	0.3204	0.2620	0.3275	1.0000	–	–	–
Price_to_Sales	0.6821	0.8008	0.6600	0.0322	1.0000	–	–
EV_to_Sales	0.6573	0.7830	0.6509	0.0248	0.9989	1.0000	–
Beta	0.4635	0.3101	0.5964	0.4294	0.3080	0.2862	1.0000

Table 11: Correlation matrix to assess multicollinearity.

Conclusions

Our analysis—combining detailed DCF projections, Monte Carlo simulations and relative-valuation regressions—consistently indicates that Arm's current market price exceeds its intrinsic value under all plausible scenarios. Even under optimistic 7-year high-growth assumptions (OM = 50%) and a conservative convergence to 10% WACC, the median DCF outcome (45.43$/share) falls well below today's trading price. Relative multiples (P/S and EV/S) calibrated on peers similarly imply fair values in the low-20s, suggesting a premium of roughly 75%. While optionality around manufacturing and AI-driven upsides could justify some uplift, these effects appear insufficient to bridge the gap to market levels. Arm looks richly valued relative to both its own cash-flow fundamentals and the broader semiconductor peer group.

Accordingly, our recommendation is to **sell** in both the 5-year and 7-year scenarios, although we discourage short-selling given the stock's high volatility and the general increase in sector market capitalizations.

References

[1] Q3 FYE25 Investor Presentation

[2] Form 20-F FYE24

[3] Form 6-K Q4 FY25 – **unaudited**

[4] Python code – GitHub

Appendix

A: Data Source and Audit Status

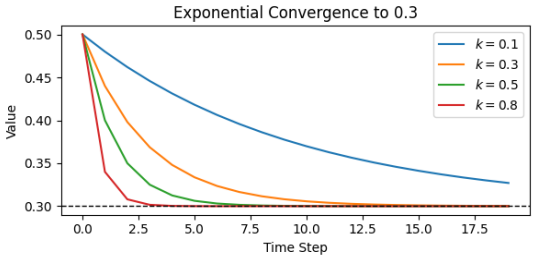
The new 20-F filing is expected around June or July, even though the last quarter of fiscal year 2025 will be reported in May. Therefore, the most recent data used have been taken from the latest 6-K filing, which is unaudited. However, it does not include all necessary information; for instance, the revenue by geographical region, which is used for the ERP calculation, is available only in the 20-F.

B: Convergence Formula

To model convergence over time, we use the recursive formula:

$$x_t = x_{t-1} + k \cdot (x^* - x_{t-1})$$

where x_t is the value at time t, x^* is the target value, and $k \in (0, 1]$ is the convergence rate. This creates exponential convergence: the gap between x_t and x^* shrinks geometrically with each step. In the course, it was referred to as the "distance formula" and a convergence rate of $k = 0.5$ was used throughout the analysis.



C: Calculation of RD fixed OM 0.5 7y

Year	CapRD	w_0	w_1	w_2	w_9	w_10	rd_lag_0	rd_lag_1	rd_lag_2	rd_lag_3	rd_lag_4	rd_lag_5	rd_lag_6	rd_lag_7	rd_lag_8	rd_lag_9	rd_lag_10
2025	5.9434	1.0	0.9	0.8	0.1	0.0	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2026	6.4162	1.0	0.9	0.8	0.1	0.0	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000	0.0000	0.0000
2027	6.9711	1.0	0.9	0.8	0.1	0.0	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000	0.0000
2028	7.6220	1.0	0.9	0.8	0.1	0.0	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000
2029	8.3852	1.0	0.9	0.8	0.1	0.0	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000
2030	9.2798	1.0	0.9	0.8	0.1	0.0	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000
2031	10.3281	1.0	0.9	0.8	0.1	0.0	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140
2032	11.6377	1.0	0.9	0.8	0.1	0.0	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950
2033	11.7364	1.0	0.9	0.8	0.1	0.0	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330
2034	11.1835	1.0	0.9	0.8	0.1	0.0	1.4315	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790
2035	10.3914	1.0	0.9	0.8	0.1	0.0	1.1376	1.4315	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710
2036	9.3467	1.0	0.9	0.8	0.1	0.0	0.7916	1.1376	1.4315	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720

Year	Amort	w_0	w_1	w_2	w_9	w_10	rd_lag_0	rd_lag_1	rd_lag_2	rd_lag_3	rd_lag_4	rd_lag_5	rd_lag_6	rd_lag_7	rd_lag_8	rd_lag_9	rd_lag_10
2025	0.4921	0.0	0.1	0.1	0.1	0.1	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2026	0.6992	0.0	0.1	0.1	0.1	0.1	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000	0.0000	0.0000
2027	0.8164	0.0	0.1	0.1	0.1	0.1	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000	0.0000
2028	0.9535	0.0	0.1	0.1	0.1	0.1	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000	0.0000
2029	1.1140	0.0	0.1	0.1	0.1	0.1	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000	0.0000
2030	1.3017	0.0	0.1	0.1	0.1	0.1	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140	0.0000
2031	1.5213	0.0	0.1	0.1	0.1	0.1	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950	0.8140
2032	1.6969	0.0	0.1	0.1	0.1	0.1	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330	0.9950
2033	1.8980	0.0	0.1	0.1	0.1	0.1	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790	1.1330
2034	1.9844	0.0	0.1	0.1	0.1	0.1	1.4315	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710	1.9790
2035	1.9297	0.0	0.1	0.1	0.1	0.1	1.1376	1.4315	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720	2.0710
2036	1.8363	0.0	0.1	0.1	0.1	0.1	0.7916	1.1376	1.4315	1.9967	3.0065	2.5697	2.1963	1.8772	1.6044	1.3713	1.1720

Glossary

Multicollinearity Strong correlation among two or more predictors in a regression model, leading to unstable coefficient estimates and inflated standard errors.

Heteroscedasticity Non-constant variance of the regression residuals across levels of an explanatory variable, which makes OLS estimates inefficient and inference unreliable.

Homoscedasticity Constant variance of the regression residuals across all levels of the explanatory variables, a key Gauss–Markov assumption ensuring OLS is the best linear unbiased estimator.